Embryo Stage Classification using Deep Learning with a Custom Ordinal Loss Function

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This work addresses the problem of automated embryo stage classification from time-lapse microscopy images using deep learning. We train four pretrained convolutional neural networks — **MobileNetV2**, **VGG16**, **VGG19**, and **InceptionV3** — on the publicly available Embryo Dataset [**embryo_dataset**], and introduce a **custom total loss function** that combines standard Cross-Entropy with an ordinal penalty term designed specifically for the sequential nature of embryonic development.

Dataset. The Embryo Dataset contains time-lapse images of human embryos annotated with 16 developmental phases: `tPB2, tPNa, tPNf, t2, t3, t4, t5, t6, t7, t8, t9+, tM, tSB, tB, tEB, tHB`. The dataset comprises 52,339 frames sampled at a rate of 1 frame per 6 across 704 embryo recordings. The class distribution is heavily imbalanced, as shown in Table 1. To prevent data leakage, splits are performed at the embryo level: 70% training (36,637 samples), 15% validation (7,932 samples), and 15% test (7,770 samples). Balanced class weights computed via scikit-learn are applied to the loss function to mitigate the imbalance, particularly for class 15 (`tHB`), which contains only 17 samples.

Table 1: Class distribution across all 16 embryo development phases.

Label	Phase	Count	Label	Phase	Count
0	tPB2	1,672	8	t7	1,926
1	tPNa	7,392	9	t8	5,593
2	tPNf	1,366	10	t9+	8,681
3	t2	5,062	11	tM	3,014
4	t3	1,031	12	tSB	3,041
5	t4	5,053	13	tB	1,910
6	t5	1,532	14	tEB	3,470
7	t6	1,579	15	tHB	17

Preprocessing and Augmentation. Images are resized to 224×224 pixels for MobileNetV2, VGG16, and VGG19, and to 299×299 for InceptionV3. Training augmentations include random cropping, horizontal flipping, rotation (± 15), and colour jitter (brightness, contrast, saturation). All images are normalised using ImageNet mean and standard deviation. The dataset's nested folder structure (some embryos store frames in subdirectories such as `F0/`) is handled by a recursive image collector that walks the full directory tree.

Model Architecture. All four models are initialised with ImageNet pretrained weights and their classification heads are replaced to output 16 classes:

- **MobileNetV2**: the `classifier` head is replaced with `Dropout(0.3)` followed by a `Linear` layer.
- **VGG16** / **VGG19**: `classifier[6]` is replaced with a `Linear(4096, 16)` layer.

- **InceptionV3**: both the auxiliary head (`AuxLogits.fc`) and the main head (`fc`) are replaced with `Linear` layers outputting 16 classes.

Custom Total Loss Function. Embryo development is inherently **sequential** — confusing adjacent stages (e.g. $t4$ with $t5$) should be penalised less than skipping multiple stages (e.g. $t4$ with tM). Standard Cross-Entropy treats all misclassifications equally. We therefore define a custom total loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \alpha \cdot \mathcal{L}_{\text{ord}}, \quad \alpha = 0.5, \tag{1}$$

where $\mathcal{L}_{\text{CE}}$ is the standard weighted Cross-Entropy loss (equivalent to Sparse Categorical Cross-Entropy), and $\mathcal{L}_{\text{ord}}$ is the **Ordinal Penalty Loss**:

$$\mathcal{L}_{\text{ord}}(\mathbf{p}, y) = \frac{1}{(C-1)^2} \sum_{k=0}^{C-1} p_k \cdot (y-k)^2, \tag{2}$$

where $C = 16$ is the number of classes, $y \in \{0, \ldots, C-1\}$ is the true label, and $p_k = \text{softmax}(\mathbf{z})_k$ is the predicted probability for class k. The denominator $(C-1)^2$ normalises the term to be scale-compatible with $\mathcal{L}_{\text{CE}}$.

The loss in Eq. (2) satisfies the following desirable properties:

1. **Non-negativity**: Since $p_k \geq 0$ and $(y-k)^2 \geq 0$ for all k, the sum is always non-negative.

2. **Zero at the optimum**: When the model assigns all probability mass to the correct class ($p_y = 1$, $p_{k \neq y} = 0$), every term in Eq. (2) vanishes, giving $\mathcal{L}_{\text{ord}} = 0$.

3. **Differentiability**: The composition of softmax (smooth) and a quadratic function is smooth everywhere, making the loss fully differentiable and compatible with gradient-based optimisation.

4. **Ordinal awareness**: The squared distance $(y-k)^2$ grows with the developmental stage gap between the true class y and any predicted class k, so skipping multiple stages incurs a strictly larger penalty than adjacent misclassification.

5. **Scale compatibility**: Dividing by $(C-1)^2$ keeps $\mathcal{L}_{\text{ord}}$ in the same order of magnitude as $\mathcal{L}_{\text{CE}}$, preventing one term from dominating.

Training Protocol. All models are trained with the Adam optimiser (learning rate 2×10^{-4}, weight decay 10^{-4}) and cosine annealing scheduling over 5 epochs with a minimum rate of 10^{-6}. Gradient clipping (max norm 2.0) is applied at every step. Early stopping with patience 3 is used, restoring the best checkpoint (selected by validation macro-F1) for final test evaluation. Each model is trained twice — once with the baseline Cross-Entropy loss and once with the custom total loss in Eq. (1) — to enable a direct comparison. For InceptionV3 an auxiliary loss term (weighted 0.4) is added during training to regularise the deeper network.

Results. Table 2 reports test accuracy and macro-F1 for both loss configurations across all four architectures.

Table 2: Test accuracy and macro-F1 — Baseline CE vs. Custom Loss (CE + Ordinal). ΔAcc denotes the improvement of the custom loss over the baseline.

Model	**Base Acc**	**Base F1**	**Custom Acc**	**Custom F1**	**ΔAcc**
MobileNetV2	0.3035	0.3177	0.3181	0.3292	+0.0146
VGG16	0.2731	0.2785	0.2257	0.2364	−0.0474
VGG19	0.2633	0.2624	0.2482	0.2453	−0.0151
InceptionV3	0.3119	0.3124	–	–	–

MobileNetV2 achieves the best performance under the custom loss (Acc: 0.3181, F1: 0.3292), demonstrating that the ordinal penalty provides a measurable benefit for this architecture. The relatively modest accuracies across all models reflect the difficulty of the 16-class fine-grained classification problem, the extreme class imbalance (class 15 has only 17 training samples), and the limited number of training epochs (5) used in this study.

Conclusion. We presented a framework for embryo stage classification using four CNN architectures and a custom total loss that augments standard Cross-Entropy with an ordinal penalty. The ordinal penalty is theoretically motivated by the sequential nature of embryonic development and satisfies five desirable properties: non-negativity, zero at the optimum, differentiability, ordinal awareness, and scale compatibility. MobileNetV2 trained with the custom loss achieves the best test accuracy (0.3181) and macro-F1 (0.3292). Future work will explore longer training schedules, progressive unfreezing of pretrained layers, and alternative ordinal formulations such as Earth Mover's Distance loss.